coop-careers-data-proj-1

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Initial Setup

```r
# load libraries
library(tidyverse)
library(lubridate)
library(tidymodels)
library(writexl)
```

Data Load

```r
# load raw data
data <- read.csv("raw_data/nyc_restaurant_inspection.csv")
```

```r
# peek at columns, what their values are
glimpse(data)
```

```
## Rows: 319,818
## Columns: 26
## $ CAMIS                 <int> 40557912, 50016943, 50060883, 40393093, 50018441~
## $ DBA                   <chr> "KISMAT INDIAN CUISINE", "EL NUEVO ROBLE BILLIAR~
## $ BORO                  <chr> "Manhattan", "Bronx", "Manhattan", "Queens", "Qu~
## $ BUILDING              <chr> "603", "216", "105A", "9215", "13311", "22", "26~
## $ STREET                <chr> "FORT WASHINGTON AVENUE", "EAST  170 STREET", "D~
## $ ZIPCODE               <chr> "10040", "10456", "10040", "11435", "11417", "10~
## $ PHONE                 <chr> "2127958633", "3478206981", "6468526499", "71852~
## $ CUISINE.DESCRIPTION   <chr> "Indian", "Spanish", "Spanish", "Latin American"~
## $ INSPECTION.DATE       <chr> "03/05/2018", "06/08/2019", "08/28/2018", "04/29~
## $ ACTION                <chr> "Violations were cited in the following area(s).~
## $ VIOLATION.CODE        <chr> "10F", "04L", "10F", "10B", "06D", "08C", "10I",~
## $ VIOLATION.DESCRIPTION <chr> "Non-food contact surface improperly constructed~
## $ CRITICAL.FLAG         <chr> "Not Critical", "Critical", "Not Critical", "Not~
## $ SCORE                 <int> 17, 22, 12, 22, 18, 12, 9, 37, 27, 9, 11, 21, 36~
## $ GRADE                 <chr> "", "B", "", "", "", "A", "A", "Z", "", "A", "A"~
## $ GRADE.DATE            <chr> "", "06/08/2019", "", "", "", "08/06/2019", "02/~
## $ RECORD.DATE           <chr> "04/03/2022", "04/03/2022", "04/03/2022", "04/03~
## $ INSPECTION.TYPE       <chr> "Cycle Inspection / Initial Inspection", "Cycle ~
```

```
## $ Latitude              <dbl> 40.85523, 40.83839, 40.86187, 40.70170, 40.67523~
## $ Longitude             <dbl> -73.93685, -73.91294, -73.92491, -73.80544, -73.~
## $ Community.Board       <int> 112, 204, 112, 412, 410, 103, 402, 103, 311, 108~
## $ Council.District      <int> 10, 16, 10, 27, 32, 1, 26, 2, 47, 4, 49, 32, 45,~
## $ Census.Tract          <int> 27300, 17902, 29100, 21200, 8600, 2900, 100, 420~
## $ BIN                   <int> 1084212, 2007983, 1064719, 4213988, 4248035, 100~
## $ BBL                   <dbl> 1021797501, 2028310056, 1022170051, 4099970001, ~
## $ NTA                   <chr> "MN35", "BX14", "MN01", "QN61", "QN56", "MN27", ~
```

Data Cleaning

Ok, so what does **drop_na()** do in R? From direct documentation: drops rows where any column (in this case, the entire dataset/dataframe) contains a missing value. So, in this case, anything with NA.

wtf is NA?

NA is different from "empty"; **NA** is considered "separate" and "missing", **"empty"** or **" "** is a separate thing completely.

So first, we need to add row numbers; this is pre-processing for later.

```r
# throwing a row number on the original data so we can track stuff later
data_with_id <- data |>
  mutate(original_row = row_number())
```

My prof from undergrad described pivot_longer as this: *"For when some of the column names are not names of variables, but values of a variable"*

```r
# we want to find where "NA" values are
na_locations <- data_with_id |>
  # first, just grab the rows that have an NA somewhere in them
  filter(if_any(-original_row, is.na)) |>
  # make everything a character so the pivot doesn't give us errors
  mutate(across(-original_row, as.character)) |>
  # convert our table to a long/increase rows and
  pivot_longer(
    cols = -original_row,
    names_to = "column_name",
    values_to = "value"
  ) |>
  # now, just keep the lines that are NA
  filter(is.na(value))

# look at list
print(na_locations)
```

```
## # A tibble: 40,326 x 3
##    original_row column_name      value
##           <int> <chr>            <chr>
##  1           16 SCORE            <NA>
##  2           17 SCORE            <NA>
```

```
## 3           22 Community.Board  <NA>
## 4           22 Council.District <NA>
## 5           22 Census.Tract     <NA>
## 6           22 BIN              <NA>
## 7           36 SCORE            <NA>
## 8           63 SCORE            <NA>
## 9           64 Community.Board  <NA>
## 10          64 Council.District <NA>
## # i 40,316 more rows
```

Correct Data Types

However, we also have empty columns, so we can find that too.

```r
# ok, same thing but for the empty "" values
empty_string_locations <- data_with_id |>
  # grab any row that has an empty "" in a character column
  filter(if_any(where(is.character),function(x) x == "")) |>
  # again, making everything a character to avoid that pivot error
  mutate(across(-original_row, as.character)) |>
  # convert table into "longer" table
  pivot_longer(
    cols = -original_row,
    names_to = "column_name",
    values_to = "value"
  ) |>
  # just find the rows that are actually empty strings
  filter(value == "")

# print out list of empty value locations
print(empty_string_locations)
```

```
## # A tibble: 359,593 x 3
##    original_row column_name value
##           <int> <chr>       <chr>
## 1            1 GRADE       ""
## 2            1 GRADE.DATE  ""
## 3            3 GRADE       ""
## 4            3 GRADE.DATE  ""
## 5            4 GRADE       ""
## 6            4 GRADE.DATE  ""
## 7            5 GRADE       ""
## 8            5 GRADE.DATE  ""
## 9            9 GRADE       ""
## 10           9 GRADE.DATE  ""
## # i 359,583 more rows
```

```r
# actually clean the data here, with the correct order of operations
cleaned_data <- data |>
  # make the date columns "actual dates" first
  mutate(
    INSPECTION.DATE = mdy(INSPECTION.DATE),
```

```r
    GRADE.DATE = mdy(GRADE.DATE),
    RECORD.DATE = mdy(RECORD.DATE)
  ) |>
  # now, get rid of rows with NA and empty strings
  # why? mdy() will automatically convert "" into an NA, then drop_NA() will remove it/drop the row
  drop_na() |>
  filter(!if_any(where(is.character), function(x) x == ""))
```

```r
# quick check to see if our cleaned data looks right
glimpse(cleaned_data)
```

```
## Rows: 151,682
## Columns: 26
## $ CAMIS                 <int> 50016943, 50013774, 50078976, 50106489, 40388672~
## $ DBA                   <chr> "EL NUEVO ROBLE BILLIARDS", "PINGS", "YARDS BAR ~
## $ BORO                  <chr> "Bronx", "Manhattan", "Queens", "Manhattan", "Ma~
## $ BUILDING              <chr> "216", "22", "2632", "84", "1090", "1815", "1030~
## $ STREET                <chr> "EAST  170 STREET", "MOTT STREET", "SKILLMAN AVE~
## $ ZIPCODE               <chr> "10456", "10013", "11101", "10003", "10028", "10~
## $ PHONE                 <chr> "3478206981", "2126029988", "9179751495", "91740~
## $ CUISINE.DESCRIPTION   <chr> "Spanish", "Seafood", "American", "Chinese", "Am~
## $ INSPECTION.DATE       <date> 2019-06-08, 2019-08-06, 2022-02-15, 2022-01-26,~
## $ ACTION                <chr> "Violations were cited in the following area(s).~
## $ VIOLATION.CODE        <chr> "04L", "08C", "10I", "04L", "08A", "06D", "08A",~
## $ VIOLATION.DESCRIPTION <chr> "Evidence of mice or live mice present in facili~
## $ CRITICAL.FLAG         <chr> "Critical", "Not Critical", "Not Critical", "Cri~
## $ SCORE                 <int> 22, 12, 9, 37, 9, 11, 21, 14, 22, 9, 10, 12, 3, ~
## $ GRADE                 <chr> "B", "A", "A", "Z", "A", "A", "B", "B", "B", "A"~
## $ GRADE.DATE            <date> 2019-06-08, 2019-08-06, 2022-02-15, 2022-01-26,~
## $ RECORD.DATE           <date> 2022-04-03, 2022-04-03, 2022-04-03, 2022-04-03,~
## $ INSPECTION.TYPE       <chr> "Cycle Inspection / Re-inspection", "Cycle Inspe~
## $ Latitude              <dbl> 40.83839, 40.71431, 40.74405, 40.73102, 40.77813~
## $ Longitude             <dbl> -73.91294, -73.99876, -73.94120, -73.98981, -73.~
## $ Community.Board       <int> 204, 103, 402, 103, 108, 501, 410, 103, 404, 310~
## $ Council.District      <int> 16, 1, 26, 2, 4, 49, 32, 2, 25, 38, 10, 24, 1, 1~
## $ Census.Tract          <int> 17902, 2900, 100, 4200, 14200, 23900, 9400, 2800~
## $ BIN                   <int> 2007983, 1001689, 4000738, 1077211, 1046733, 502~
## $ BBL                   <dbl> 2028310056, 1001620012, 4000960010, 1005550021, ~
## $ NTA                   <chr> "BX14", "MN27", "QN31", "MN22", "MN40", "SI12", ~
```

```r
# summary(cleaned_data)
```

Write to CSV, XLSX

```r
# save data
write_csv(cleaned_data, "cleaned_nyc_inspections.csv")
write_xlsx(cleaned_data, "cleaned_nyc_inspections.xlsx")
```